12.17 Banach's Matchboxes

WORKSHEET

The question

A taxi driver keeps a packet of toffees and a packet of mints in her taxi. From time to time she takes either a toffee (with probability p) or mint (with probability $q = 1 - p$). At the beginning of the week she has n toffees and m mints in the packets. On the Nth occasion that she reaches for a sweet, she discovers (for the first time) that she has run out of that kind of sweet. What is the probability that she was reaching for a toffee?

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself For the driver to discover on the Nth reach that the toffees have run out, what must the Nth reach be? What must the previous $N-1$ reaches contain?

STAGE 1 Discovering the toffees are gone

Ask yourself In what order can the first $N-1$ reaches occur? What condition on $N-1-n$ ensures she never found the mint packet empty?

STAGE 3 The conditional probability

Ask yourself We are told that a discovery happens on the Nth reach. What is the probability that it was a toffee?

Pause and try Check the formula when $n=m$ and $p=q=\frac{1}{2}$. By symmetry the answer should be $\frac{1}{2}$ for every possible N. Is it?

Try this next

1. **Small case.** With $n=m=1$ and $p=\frac{1}{2}$, list the possible sequences and find the distribution of N.
2. **Banach's formula.** Verify that $\sum_{k=0}^{n}\binom{2n-k}{n}2^{-(2n-k)}=1$ for $n=1$ and $n=2$.
3. **Equal tops.** Show that $\mathrm{P}(K=0)=\mathrm{P}(K=1)$ for every n in Banach's problem.
4. **Which runs out first?** For the taxi driver, find the probability that the toffees are the first packet to be discovered empty (at any time), when $n=m=2$ and $p=\frac{2}{3}$.

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter VI, which introduced Banach's problem to the world.
- **R. Daniel Mauldin (ed.), *The Scottish Book*** (Birkhäuser). The problems from the Scottish Café, with commentary.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University

Press), on the negative binomial distribution.

The full guided solution, and what the question leads to at university, is in Section 12.17:
https://beyondstep.keithmcnulty.org/ch12/matchboxes.html